

Trainings ▾

General Information

The air intake system consists of the aftercoolers, the intercoolers, the turbochargers, the intake manifolds, and the air piping.

The QSK45 and QSK60 engines utilize outboard aftercoolers similar to KV products. The primary differences are:

- The revised inlet plumbing from the turbocharger
- Connection flange between the aftercoolers on the common bank.

These differences will **not** impact the engine's installation. Front aftercoolers incorporate a vent connection to improve the initial coolant fill; rear aftercoolers have petcocks.

The QSK60 two stage engines have incorporated an intercooler (one per bank) into the air handling system that provides for reduced air intake temperatures for improved altitude capability, fuel economy, durability, and overall performance. This two-pass core increases the installed height of the engine both at the front and rear. Along with the four additional low-pressure turbochargers, the intercooler cores will require additional space for QSK60 installations.

QSK45 and QSK60 engines use high-efficiency Holset® turbochargers. The QSK45 engines use single stage turbocharging, whereas the QSK60 has either one- or two stage turbocharging.

The QSK45-C engines have one turbocharger per engine bank and offer, as an option, either a front-out or rear-out exhaust outlet connection.

The QSK60-C (single stage) engines have two turbochargers per bank (one per aftercooler) and either a top-out or side-out exhaust outlet connection as options. The air inlet connections for the QSK60-C (single stage) are at four separate locations, two at front (right and left bank) and two at rear (right and left bank).

The QSK60 two stage engine configuration uses a two stage approach to turbocharging. The QSK60 two stage configuration uses eight turbochargers and two common intercoolers (one for the front cylinders and another for the rearmost cylinders). The intake air is first compressed in the low-pressure turbocharger. Then air is cooled through an intercooler. The second compression occurs in the high-pressure turbocharger. After the second compression, the air enters the aftercooler. The two stage turbocharging arrangement has two 45-degree exhaust outlets.

All QSK45 and QSK60 turbochargers (except low-pressure turbochargers in the QSK60 two stage engine) operate with a water-cooled turbocharger bearing housing to prolong the life of the turbocharger bearings and seals. Plumbing to the bearing housing is contained within the engine

envelope and does **not** require customer installation or adjustment. However, the turbocharger coolant plumbing requires venting upon initial engine fill.

The correct turbocharger assembly **must** be installed. Many turbochargers appear the same physically but contain different internal parts. If the wrong turbocharger assembly is installed, the performance of the engine will be less than specification, or the engine will be damaged due to excessive pressure and temperature in the cylinders. One of the dataplates that is attached to the turbocharger identifies the vendor, the turbocharger model, and the Cummins part number. Cummins Inc., identifies the turbocharger by the part number, **not** the model nomenclature.

The QSK45 exhaust outlet connections are 140 mm [5.5 in] in diameter and available as a 90 degree 125 pound American National Standards Institute flange or a 90-degree slip-fit connection for customer clamping.

The QSK60 single stage has two 230 mm [9 in] inner diameter outlets for either top-out or side-out orientations. The exhaust outlet connection flange has eight 18 mm [0.71 in] diameter holes on a 284 mm [11 in] diameter bolt circle.

The QSK60 two stage utilizes two 254 mm [10.0 in], 45 degree exhaust outlet connections. The exhaust outlet connection flanges have twelve 19 mm [0.75 in] diameter holes on a 362.0 mm [14.25 in] diameter bolt circle.

HX83 Turbocharger

The Holset® HX83 turbochargers are fitted with titanium impeller wheels, which require a post assembly balancing process.

These turbochargers are dynamically balanced using a core balancing process. This process balances the compressor wheel with the turbine wheel when it is installed into the bearing housing. This balanced assembly is called the “center housing rotating assembly”. Once completed the remainder of the turbocharger is assembled. Rebuilding the turbocharger outside of the Holset® manufacturing plant using alignment marks on the compressor wheel and turbine shaft does **not** guarantee the balance level will comply with the original manufacturing limits. This will result in an increased risk of turbocharger failure and turbocharger noise. Holset® recommends that the turbochargers be returned to Cummins Recon for repair and balancing. An alternative for rebuild is the replacement of the center housing rotating assembly. The center housing rotating assembly replacement **only** requires the removing of the compressor and turbine housings from the original turbocharger and installing the reconditioned center housing rotating assembly.

Aftercooler Core

A change has been made to the QSK60 aftercoolers to address a cracking issue with the aftercooler flange. The change in the aftercooler core design also led to changes to the intake manifold and aftercooler cover to accommodate the new core.

- Relocation of the aftercooler core capscrew mounting holes inboard of the flange-forming notch. This is similar to the QSK45, K38 and K50 aftercooler cores.

- The flange thickness was increased by 30 percent. The previous flange was 3.43 mm [0.135 in], the new flange is 4.55 mm [0.179 in].

The updated aftercooler core went into production at the Daventry engine plant with engine serial number First 33150822.

The air intake manifold and aftercooler cover were modified to accept the new aftercooler core. The new intake manifold will **not** accept the previous aftercooler core. The previous manifold and cover will **not** accept the new aftercooler core without machining. If a new cover is fitted to an previous intake manifold, then a longer roll pin is required to align the cover.

Aftercooler Compatibility Table						
Component	Previous Aftercooler Core	Updated Aftercooler Core	Previous Air Intake Manifold	Updated Air Intake Manifold	Previous Aftercooler Cover	Updated Aftercooler Cover
Previous aftercooler core			Y	N	Y	Y
Updated aftercooler core			M	Y	M	Y
Previous air intake manifold	Y	M			Y	D
Updated air intake manifold	N	Y			N	Y
Previous aftercooler cover	Y	M	Y	N		
Updated aftercooler cover	Y	Y	D	Y		
Clamping plate	Y	N	Y	N	Y	Y

The key for the aftercooler compatibility table is listed below:

Y = Yes, will fit modifications are not required

N = No, will not fit

M = Will fit after modification

D = Longer dowel pins are required to align the aftercooler cover and intake manifold.

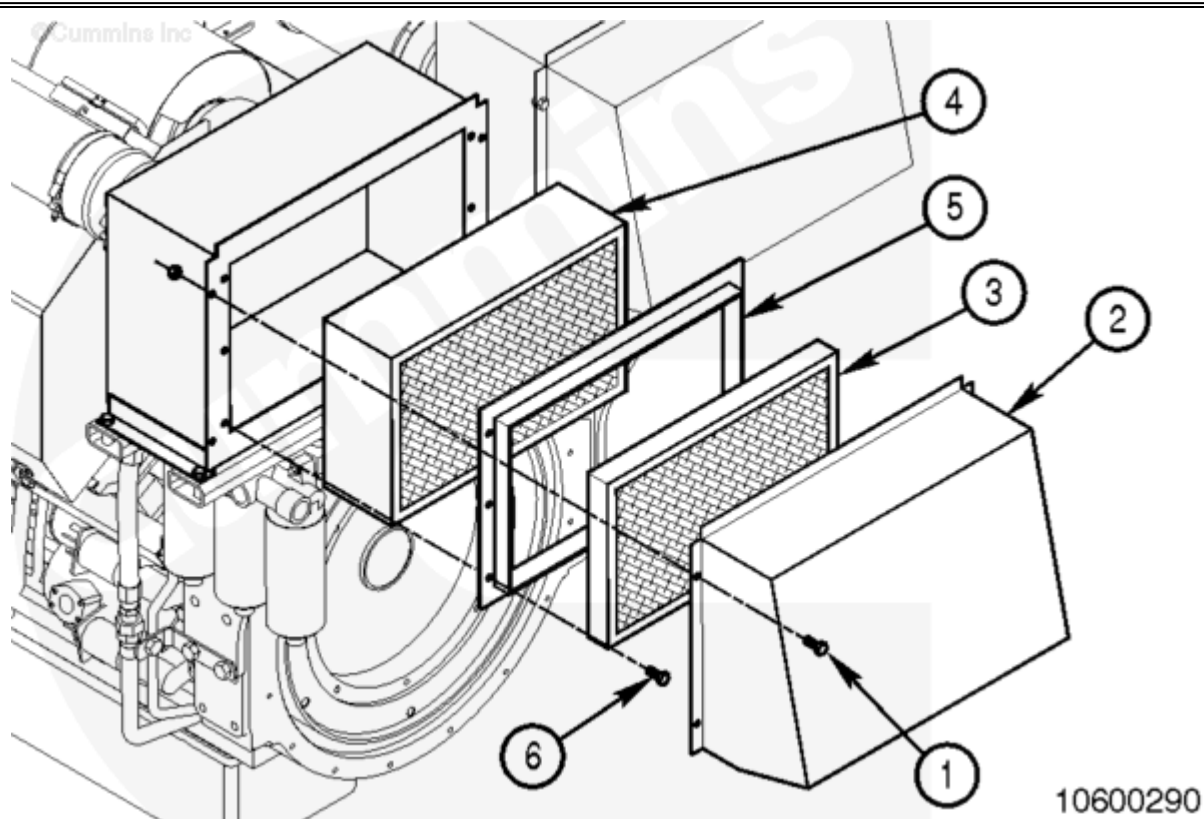
Cast Iron

With the introduction of engines with electronically actuated injectors, all QSK60's have intake manifolds and aftercooler covers that are made of cast iron. Previous engines have aluminum aftercooler assemblies. The cast iron was selected for its increased strength and durability. Use caution when removing these components, as the cast iron is heavier than aluminum. Noted also, is that grade 10.9 hardware and a higher torque value is required during installation of the cast iron components.

Marine Applications

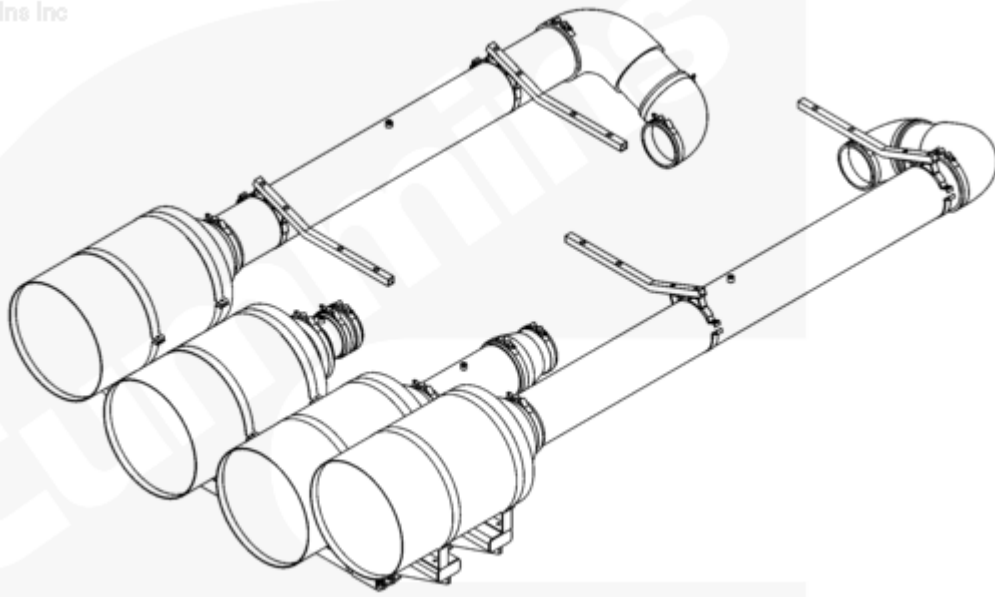
The QSK60M uses a unique engine-mounted air filter arrangement. The arrangement is mounted on a structure at the rear of the engine.

The original air cleaner assembly contains a washable prefilter (3), a replaceable filter element (4), and a restriction indicator. There is one assembly for each bank and each provides air for two turbochargers.



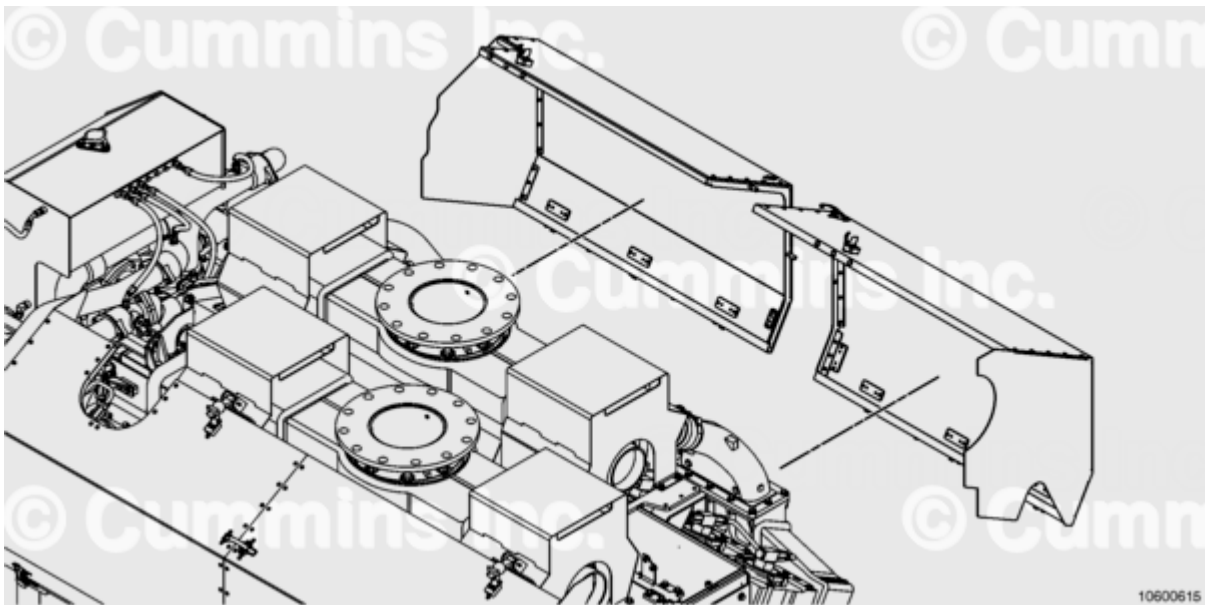
Air Cleaner Assembly

The present design uses four Fleetguard® disposable filter elements. Each one feeds a single turbocharger. Individual restriction indicators are included in the tubing from each filter.



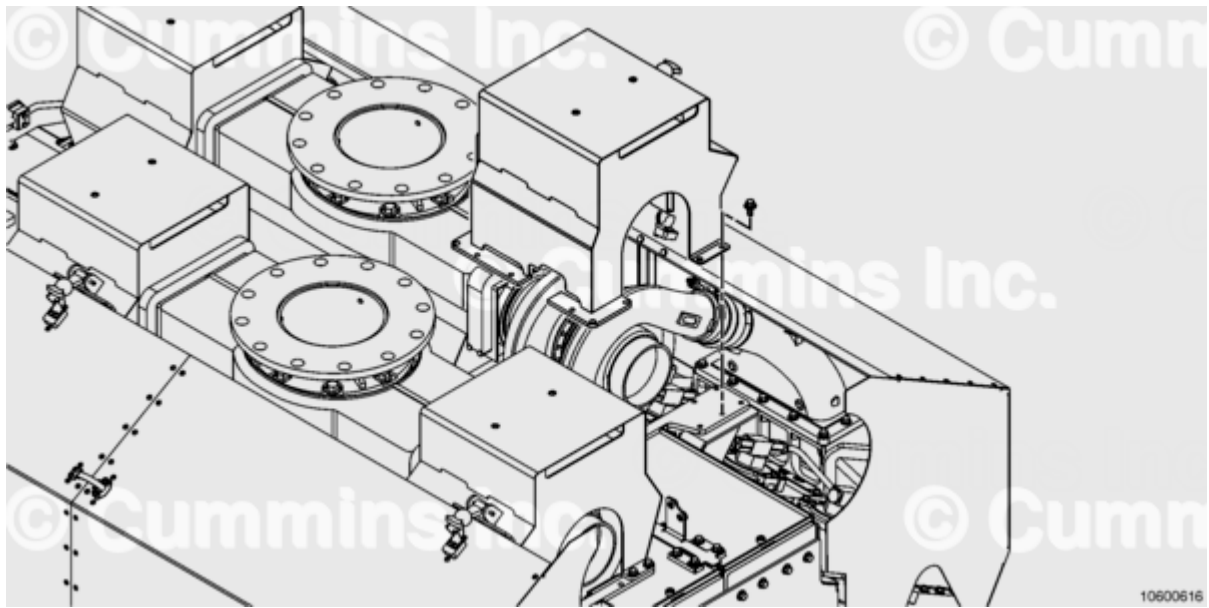
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Because of the difference in marine agency requirements, the aftercoolers, along with their inlet plumbing, **must** be shielded to keep the surface temperatures down. Each bank has two heat shields.



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The turbochargers are also covered with dry heat shields.



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